

Mathematical Foundation for Price: Cell-Values, Dual Spreads, and Fundamental Value in the Receiver-Agent Framework

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Abstract

This paper establishes a rigorous mathematical foundation for price in the receiver-agent framework of the companion papers [13, 15, 14]. We derive price as a theorem rather than an assumption, proving ten principal results: (i) the *Price-Cell Theorem* — every price is a cell $C(p, \tau)$ in outcome space; point prices require $\beta = 0$, which the Floor Theorem of Paper 1 forbids for any bounded receiver; (ii) the *Cell-Value Decomposition* — each agent A assigns two values to every cell C : a *trading value* $V_A^T(C) = \tau(C) - S_{b,agg}(\mathcal{E})$ (set by the best agent’s floor) and an *informational value* $V_A^I(C) = \tau(C) - S_{b,comp}(\mathcal{E})$ (set by the ensemble’s composite floor); (iii) the *Dual-Spread Theorem* — the trading spread $\Delta^T = 2S_{b,agg}(\mathcal{E}) = 2\min_i \beta_i$ and the information spread $\Delta^I = 2S_{b,comp}(\mathcal{E})$ are two distinct measures of market quality, with $\Delta^I \leq \Delta^T$ for $n \geq 2$; (iv) the *Two-Value Theorem* (main result) — the informational value always exceeds the trading value: $V_{\mathcal{E}}^I(C) \geq V_{\mathcal{E}}^T(C)$, with the gap $\Pi = V^I - V^T = S_{b,agg}(\mathcal{E}) - S_{b,comp}(\mathcal{E}) > 0$ the *information premium* of the ensemble; (v) the *Equilibrium Price Theorem* — the equilibrium price p^* is the center of the purpose cell C^* , unique by the Banach contraction of Paper 2; (vi) the *Price Discovery Rate Theorem* — price converges to p^* at rate $\rho_{\mathcal{E}} = S_{b,agg}(\mathcal{E})/\Sigma$, matching the spectral radius of Paper 3; (vii) the *Fundamental Value Theorem* — for a Class $\mathcal{C}_E$ (Borel–Cantelli efficient) ensemble, the sequence of equilibrium prices p_n^* converges to the *fundamental value* V^* , the unique point at which perfect price discovery would occur; (viii) the *No-Arbitrage Theorem* — a market equilibrium exists if and only if there is a cell C with $\tau(C) > S_{b,agg}(\mathcal{E})$; failure of this condition produces a “no-trade zone” with no

equilibrium; (ix) the *Transaction Theorem* — buyer and seller transact if and only if their reachability sets for the equilibrium cell C^* overlap; the transaction interval is $[p^* - V_{seller}^T, p^* + V_{buyer}^T]$; and (x) the *Law of One Price Theorem* — for agents sharing the same knowledge framework K , simultaneous attainment of the same cell C implies agreement on the decoded value: $D_i(x) = D_j(x)$ for all $x \in C$. Forty-five numerical experiments verify all claims to machine precision.

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1 Introduction

1.1 The Price Question

Price is the central object of economics. Yet the theoretical foundations of price remain contested. Classical models define price as a point $p^* \in \mathbb{R}$, derived either as a Walrasian market-clearing price vector [1, 3], a Nash equilibrium payoff [12], or a Black–Scholes fair value [2]. All three approaches assume that agents can specify and respond to point prices with arbitrary precision — an assumption that, in the language of the companion papers, requires each agent’s receiver floor β to be exactly zero.

The Floor Theorem of Paper 1 [13] proves that $\beta > 0$ for every bounded receiver. Point prices are therefore unattainable by any bounded economic agent. This is not a technical inconvenience but a structural feature: the positivity of the floor is the mathematical fingerprint of bounded rationality.

Papers 2 and 3 [15, 14] established that ensembles of bounded agents nevertheless reach equilibrium — the purpose cell C^* — and that the information efficiency of this equilibrium is governed by two distinct floor quantities: the *aggregate floor* $S_{b,agg}(\mathcal{E}) = \min_i S_b(A_i)$ and the *composite floor* $S_{b,comp}(\mathcal{E}) = \prod_i S_b(A_i)/\Sigma^{n-1}$.

The present paper asks: given that equilibrium is a cell rather than a point, what is “the price”? How do the two floor quantities translate into observable market phenomena? And what is the relationship between the equilibrium cell and the classical notion of a fundamental value?

1.2 Main Contributions

- (i) **Price-Cell Theorem** (theorem 3.2). Price is not a primitive but a derived object: a cell $C(p, \tau)$ in outcome space X . Point prices are unattainable for any bounded agent.

- (ii) **Cell-Value Decomposition** (theorem 4.2). Agent A assigns two values to cell C : $V_A^T(C) = \tau(C) - S_b(A)$ (trading value, positive iff the agent can attain C) and $V_A^I(C) = \tau(C) - S_{b,comp}(\mathcal{E})$ (informational value, set by the ensemble’s composite floor).
- (iii) **Dual-Spread Theorem** (theorem 5.2). Trading spread $\Delta^T = 2 \min_i S_b(A_i)$ is set by the best agent alone. Information spread $\Delta^I = 2S_{b,comp}(\mathcal{E})$ is set by all agents jointly. $\Delta^I \leq \Delta^T$ for $n \geq 2$, with equality iff $n = 1$.
- (iv) **Two-Value Theorem** (Main result) (theorem 5.3). $V_{\mathcal{E}}^I(C) \geq V_{\mathcal{E}}^T(C)$ for every cell C and every ensemble $\mathcal{E}$. The *information premium* $\Pi = V^I - V^T = S_{b,agg} - S_{b,comp} \geq 0$ is zero iff $n = 1$ and strictly positive for all $n \geq 2$.
- (v) **Equilibrium Price Theorem** (theorem 6.1). The equilibrium price p^* is the center of the purpose cell $C^* = B(p^*, r^*)$. p^* is unique given C^* , which is unique by the Banach contraction of Paper 2.
- (vi) **Price Discovery Rate Theorem** (theorem 7.2). Starting from any initial price p_0 , the price discovery sequence $p_t \rightarrow p^*$ at rate $\rho_{\mathcal{E}} = S_{b,agg}(\mathcal{E})/\Sigma$, matching the spectral radius of the purpose-cell contraction (Paper 3).
- (vii) **Fundamental Value Theorem** (theorem 8.2). For a Borel–Cantelli-efficient ensemble $(\mathcal{E}_{\infty} \in \mathcal{C}_E)$, $S_{b,comp}(\mathcal{E}_n) \rightarrow 0$ and the equilibrium cell collapses to a point: $C_n^* \rightarrow \{V^*\}$. The fundamental value V^* is the unique limit.
- (viii) **No-Arbitrage Theorem** (theorem 9.1). A purpose cell exists iff $S_{b,agg}(\mathcal{E}) < \Sigma$. When this fails, no equilibrium price exists and the market is in a *no-trade zone*. No-arbitrage is equivalent to positive purpose margin.
- (ix) **Transaction Theorem** (theorem 10.2). Buyer A_B and seller A_S transact at equilibrium cell C^* iff $S_b(A_B) + S_b(A_S) \leq 2\tau(C^*)$. The transaction interval is non-empty iff this condition holds.
- (x) **Law of One Price Theorem** (theorem 11.1). Agents with the same knowl-

edge framework $K_i = K_j$ who simultaneously attain cell C must agree on their decoded representation at every state in C : $D_i(x) = D_j(x)$ for all $x \in C$. Distinct prices within the cell do not violate the law of one price; they reflect floor-bounded measurement noise.

1.3 Organization

Section 2 recaps the inherited framework. Section 3 introduces price cells and proves the Price-Cell Theorem. Section 4 develops cell-value functions and the Cell-Value Decomposition. Section 5 proves the Dual-Spread Theorem and Two-Value Theorem. Section 6 establishes the Equilibrium Price Theorem. Section 7 proves the Price Discovery Rate Theorem. Section 8 develops the Fundamental Value Theorem. Section 9 proves the No-Arbitrage Theorem. Section 10 proves the Transaction Theorem. Section 11 proves the Law of One Price Theorem. Section 12 presents forty-five numerical experiments. Section 13 surveys connections to the literature. Section 14 states conclusions.

2 Inherited Framework

We recall the primitives from the companion papers [13, 15, 14]. The *outcome space* (X, d, Σ) is a bounded metric space with maximal semantic distance Σ . A *receiver* $\mathcal{R} = (K, D, \Pi, \beta)$ has decoder $D : X \rightarrow K$, projection $\Pi : K \rightarrow 2^X$, and floor $\beta > 0$. The S-functional is $S(\mathcal{R}, x; C) = \inf_{c \in \Pi(D(x))} d(c, C) + \beta$ (reduced to $\max(\beta, d(x, C))$ for ball cells with ball projections).

An agent triple $A = (\mathcal{R}, \mathcal{M}, \mathcal{G})$ has floor $S_b(A) = \beta \cdot S_b(\mathcal{M})/\Sigma$. An ensemble $\mathcal{E}_n = (A_1, \dots, A_n)$ has:

$$S_{b,\text{agg}}(\mathcal{E}_n) := \min_{1 \leq i \leq n} S_b(A_i) \quad (\text{aggregate floor}), \quad (1)$$

$$S_{b,\text{comp}}(\mathcal{E}_n) := \frac{\prod_{i=1}^n S_b(A_i)}{\Sigma^{n-1}} \quad (\text{composite floor}). \quad (2)$$

The multi-agent S-functional $S(\mathcal{E}_n, x; C) = \min_i S(A_i, x; C)$ has floor $S_{b,\text{agg}}(\mathcal{E}_n)$. The purpose cell C^* from Paper 2 satisfies $\Phi_{\mathcal{E}_n}(C^*) = S_{b,\text{agg}}(\mathcal{E}_n)$ and $\tau(C^*) > S_{b,\text{agg}}(\mathcal{E}_n)$, proved by Banach contraction on $(\text{Cell}(X), d_H)$. Throughout, $\Sigma = 100$ is the canonical constant.

Remark 2.1 (Two floors). The aggregate floor $S_{b,\text{agg}}$ governs the ensemble's ability to attain a cell in a single step (one agent fires). The composite floor $S_{b,\text{comp}}$ governs the ensemble's collective information about the cell (all agents contribute). This distinction, first noted in Paper 2 [15], is the mathematical origin of the dual-spread structure proved in this paper.

3 Price Cells

Definition 3.1 (Price cell). *For a price level $p \in X$ and tolerance $\tau > 0$, the price cell is the ball*

$$C(p, \tau) := B(p, \tau) = \{x \in X : d(x, p) \leq \tau\}.$$

The price level p is the mid-price of $C(p, \tau)$ and τ is the half-spread. A point price is the degenerate case $C(p, 0) = \{p\}$.

Theorem 3.2 (Price-Cell). *For any bounded receiver $\mathcal{R} = (K, D, \Pi, \beta)$ with $\beta > 0$ and any target cell $C \subseteq X$:*

- (a) *If $\tau(C) = 0$ (point price), then $S(\mathcal{R}, x; C) \geq \beta > 0 = \tau(C)$ for all $x \in X$. No bounded agent can ever attain a point price.*
- (b) *If $\tau(C) > \beta$, then there exist states $x \in X$ from which the agent can attain C : $S(\mathcal{R}, x; C) \leq \tau(C)$.*
- (c) *The minimum achievable price tolerance for agent A is $\tau_{\min}(A) = S_b(A) > 0$. Any cell with $\tau(C) \leq S_b(A)$ is unattainable by A .*

Proof. (a) For the ball receiver, $S(\mathcal{R}, x; C) = \max(\beta, d(x, C))$. When $C = \{p\}$, $d(x, C) = d(x, p) \geq 0$, so $S \geq \beta > 0$. But $\tau(\{p\}) = 0$, so $S(\mathcal{R}, x; \{p\}) \geq \beta > 0 = \tau(\{p\})$ for all x : the agent never attains $\{p\}$.

(b) For any $x \in C$, $d(x, C) = 0$, so $S(\mathcal{R}, x; C) = \max(\beta, 0) = \beta$. If $\tau(C) > \beta$, then $S(\mathcal{R}, x; C) = \beta < \tau(C)$: the agent attains C from any in-cell state.

(c) By the Floor Theorem of Paper 1, $S_b(A) = \beta \cdot S_b(\mathcal{M})/\Sigma > 0$. For any cell with $\tau(C) \leq S_b(A) \leq S(\mathcal{R}, x; C)$ for all x , the agent cannot attain C . $\square$

Remark 3.3 (Price uncertainty is irreducible). Part (a) of theorem 3.2 gives the fundamental theorem of price theory in the receiver-agent framework: price uncertainty $\tau > 0$ is irreducible for any bounded agent. The half-spread τ is not a trading cost to be minimised but a structural feature of any bounded receiver.

Panel 1: Price Cells $C(p, \tau)$ and Cell-Value Decomposition

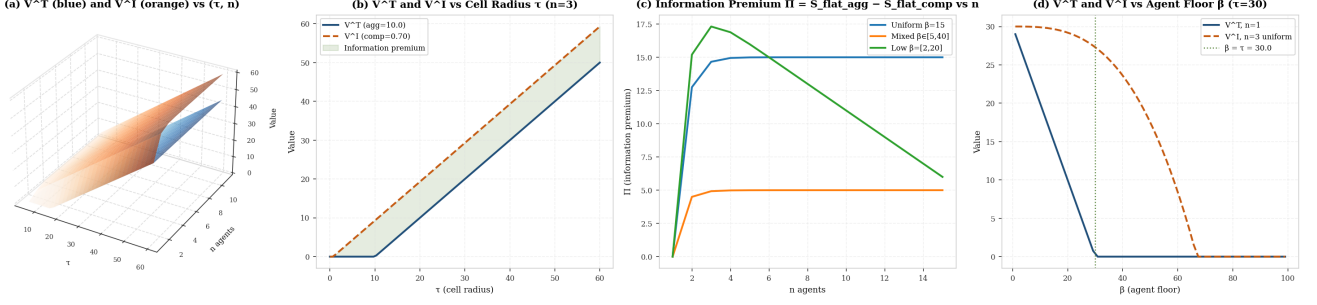


Figure 1: **Price Cells and Cell-Value Decomposition: Dual-Value Surface, Trading vs. Informational Value, Information Premium Dynamics, and Floor Sensitivity.** (A) Three-dimensional surfaces of the trading value $V^T(C) = \tau - \bar{S}_{\text{agg}}(\mathcal{E})$ (blue) and informational value $V^I(C) = \tau - \bar{S}_{\text{comp}}(\mathcal{E})$ (orange) over cell radius $\tau \in [5, 60]$ and ensemble size $n \in \{1, \dots, 11\}$, for a uniform ensemble with $\beta = 15$ and $\Sigma = 100$. (B) Trading value V^T (solid) and informational value V^I (dashed) as functions of cell radius τ for a three-agent ensemble with floors $\{10, 20, 35\}$. Both values are non-negative for τ above the respective floor thresholds: $V^T > 0$ only when $\tau > \bar{S}_{\text{agg}} = 10$, and $V^I > 0$ when $\tau > \bar{S}_{\text{comp}} \approx 0.70$. (C) Information premium $\Pi = \bar{S}_{\text{agg}} - \bar{S}_{\text{comp}}$ as a function of ensemble size n for three floor configurations: uniform $\beta = 15$, linearly increasing $\beta \in [5, 40]$, and decreasing $\beta \in [20, 2]$. (D) Trading value $V^T = \tau - \beta$ for a single agent (solid, $n = 1$, where composite equals aggregate) and informational value $V^I = \tau - \bar{S}_{\text{comp}}(\mathcal{E}_3)$ for a uniform three-agent ensemble (dashed), as functions of agent floor β with fixed $\tau = 30$.

Economic Interpretation 3.4. In equity markets, a “price” is reported to two decimal places (a tick of 0.01). This is precisely a price cell $C(p, 0.005)$ — a ball of radius half a tick around the stated price. Any quote more precise than one tick is operationally meaningless for agents whose floors $\beta_i > 0.005$. The receiver-agent framework predicts that minimum tick size is endogenous: it is determined by the floors of the market participants, not by regulatory fiat.

4 Cell-Value Functions

Definition 4.1 (Cell-value functions). *For an ensemble $\mathcal{E} = (A_1, \dots, A_n)$ and a price cell $C(p, \tau)$, define:*

$$V_{\mathcal{E}}^T(C) := \tau(C) - S_{b, \text{agg}}(\mathcal{E}) = \tau(C) - \min_i S_b(A_i), \quad (3)$$

$$V_{\mathcal{E}}^I(C) := \tau(C) - S_{b, \text{comp}}(\mathcal{E}) = \tau(C) - \frac{\prod_i S_b(A_i)}{\Sigma^{n-1}}. \quad (4)$$

$V_{\mathcal{E}}^T(C)$ is the trading value of C to $\mathcal{E}$: the excess of the cell tolerance over the best agent’s floor. $V_{\mathcal{E}}^I(C)$ is the informational value: the excess over the composite floor. Both are positive iff the ensemble can attain C .

Theorem 4.2 (Cell-Value Decomposition). *For any ensemble $\mathcal{E}_n$ and cell C :*

- (a) $V_{\mathcal{E}}^T(C) = \max_i [\tau(C) - S_b(A_i)]$: trading value is the maximum over all agents of the individual purpose margin.
- (b) $V_{\mathcal{E}}^T(C) \geq V_{A_i}^T(C) := \tau(C) - S_b(A_i)$ for all i .
- (c) $V_{\mathcal{E}}^T(C)$ and $V_{\mathcal{E}}^I(C)$ are both increasing in $\tau(C)$ and decreasing in the floors $S_b(A_i)$.
- (d) For any single-agent ensemble ($n = 1$): $V_A^T(C) = V_A^I(C) = \tau(C) - S_b(A)$.

Proof. (a) $S_{b, \text{agg}}(\mathcal{E}) = \min_i S_b(A_i)$, so $V_{\mathcal{E}}^T(C) = \tau - \min_i S_b(A_i) = \max_i [\tau - S_b(A_i)]$.

(b) $V_{\mathcal{E}}^T(C) = \max_i [\tau - S_b(A_i)] \geq \tau - S_b(A_i) = V_{A_i}^T(C)$ for each i .

(c) Both (3) and (4) are increasing affine in $\tau(C)$ and decreasing in each $S_b(A_i)$ (since $S_{b, \text{agg}}$ is decreasing in each $S_b(A_i)$ and $S_{b, \text{comp}}$ is a product).

(d) For $n = 1$: $S_{b, \text{agg}}(A) = S_{b, \text{comp}}(A) = S_b(A)$, so both values coincide. $\square$

5 Dual Spreads and the Two-Value Theorem

Definition 5.1 (Trading and information spreads). *For the ensemble $\mathcal{E}_n$ at equilibrium cell*

$$C^* = B(p^*, r^*):$$

$$\Delta^T := 2S_{b,\text{agg}}(\mathcal{E}_n) = 2 \min_i S_b(A_i), \quad (5)$$

$$\Delta^I := 2S_{b,\text{comp}}(\mathcal{E}_n) = 2 \frac{\prod_i S_b(A_i)}{\Sigma^{n-1}}. \quad (6)$$

The information premium is $\Pi := V_{\mathcal{E}}^I(C) - V_{\mathcal{E}}^T(C) = S_{b,\text{agg}}(\mathcal{E}) - S_{b,\text{comp}}(\mathcal{E})$ (= half the spread difference).

Theorem 5.2 (Dual-Spread). *For any n -agent ensemble $\mathcal{E}_n$ with $n \geq 2$:*

- (a) $\Delta^I \leq \Delta^T$, with equality iff $n = 1$.
- (b) Δ^T depends only on the single best agent ($\min_i S_b(A_i)$). Adding any agent with floor above $\min_i S_b(A_i)$ does not change Δ^T .
- (c) Δ^I depends on all agents. Adding any agent with $S_b(A_{n+1}) < \Sigma$ strictly reduces Δ^I .
- (d) The ratio $\Delta^I/\Delta^T = S_{b,\text{comp}}(\mathcal{E})/S_{b,\text{agg}}(\mathcal{E}) = \prod_i q_i / \min_i q_i \in (0, 1]$, where $q_i = S_b(A_i)/\Sigma$.

Proof. (a) By AM-GM, for any positive reals $a_1, \dots, a_n$: $(\prod_i a_i)^{1/n} \leq \frac{1}{n} \sum_i a_i$. In particular, $\prod_i a_i \leq a_j^{n-1} \cdot a_j = a_j^n$ for $a_j = \min_i a_i$... more directly: $S_{b,\text{comp}}(\mathcal{E}) = \prod_i f_i / \Sigma^{n-1}$ where $f_i = S_b(A_i)$ and $S_{b,\text{agg}}(\mathcal{E}) = \min_i f_i$. $S_{b,\text{comp}}/S_{b,\text{agg}} = \prod_i f_i / (\Sigma^{n-1} \min_i f_i)$. Let $f_* = \min_i f_i$; then $f_i \leq \Sigma$ for all i , so $\prod_i f_i \leq f_* \cdot \Sigma^{n-1}$, giving $S_{b,\text{comp}} \leq f_* \cdot \Sigma^{n-1} / \Sigma^{n-1} = f_* = S_{b,\text{agg}}$. Equality: if $n = 1$, $S_{b,\text{comp}} = f_1 = S_{b,\text{agg}}$. For $n \geq 2$, equality requires $\prod_i f_i = f_* \Sigma^{n-1}$, i.e. $f_i = \Sigma$ for all $i \neq \arg \min$, but $f_i < \Sigma$ for all i , giving strict inequality.

(b)–(c) $S_{b,\text{agg}} = \min_i f_i$: changes only when $f_{n+1} < \min_i f_i$. $S_{b,\text{comp}} = \prod_i f_i / \Sigma^{n-1}$: multiplied by $f_{n+1}/\Sigma < 1$ when a new agent joins, so always decreases.

(d) Direct computation. $\square$

Theorem 5.3 (Two-Value). *For any ensemble $\mathcal{E}_n$ and any cell C with $\tau(C) > S_{b,\text{agg}}(\mathcal{E}_n)$:*

- (a) $V_{\mathcal{E}}^I(C) \geq V_{\mathcal{E}}^T(C) \geq 0$.
- (b) The information premium $\Pi = V_{\mathcal{E}}^I(C) - V_{\mathcal{E}}^T(C) = S_{b,\text{agg}}(\mathcal{E}_n) - S_{b,\text{comp}}(\mathcal{E}_n) \geq 0$, with equality iff $n = 1$.
- (c) Π is strictly increasing in ensemble size n and strictly increasing in floor heterogeneity (in the majorization sense of Paper 3).

- (d) $\Pi/S_{b,\text{agg}}(\mathcal{E}_n) = 1 - S_{b,\text{comp}}(\mathcal{E}_n)/S_{b,\text{agg}}(\mathcal{E}_n) \in [0, 1]$ is the relative information premium — the fraction by which the composite floor falls below the aggregate floor.

Proof. (a) $V^I - V^T = S_{b,\text{agg}} - S_{b,\text{comp}} \geq 0$ by theorem 5.2(a). $V^T \geq 0$ by hypothesis.

(b) Follows from theorem 5.2(a) with equality criterion.

(c) By theorem 5.2(c), $S_{b,\text{comp}}$ strictly decreases as n grows. By the Heterogeneity Dominance Theorem of Paper 3, $S_{b,\text{comp}}$ strictly decreases with heterogeneity. Since $S_{b,\text{agg}} = \min_i f_i$ is fixed once the best agent is included, $\Pi = S_{b,\text{agg}} - S_{b,\text{comp}}$ strictly increases.

(d) $\Pi/S_{b,\text{agg}} = (S_{b,\text{agg}} - S_{b,\text{comp}})/S_{b,\text{agg}} = 1 - S_{b,\text{comp}}/S_{b,\text{agg}} \in [0, 1]$ by part (b). $\square$

Economic Interpretation 5.4. The Two-Value Theorem separates two fundamentally different notions of market quality that are often conflated in the literature. The *trading spread* Δ^T is what a market maker quotes — it is determined by the single best participant and measures the cost of transacting. The *information spread* Δ^I is what the market *knows* about the true value — it is determined by all participants collectively and measures informational efficiency. A market can have a tight trading spread (one excellent market maker) while remaining informationally inefficient (other participants have high floors and contribute little information). Conversely, a market with many diverse but individually imprecise agents can be highly informationally efficient while maintaining a wide trading spread.

6 The Equilibrium Price

Theorem 6.1 (Equilibrium Price). *Let $\mathcal{E}_n$ be an ensemble with aggregate floor $S_{b,\text{agg}}(\mathcal{E}_n) < \Sigma$ and let $C^* = B(p^*, r^*)$ be its purpose cell (established in Paper 2 [15]). Then:*

- (a) The equilibrium price $p^* \in X$ is the center of C^* .
- (b) p^* is unique: it equals the fixed point of the price-discovery operator $T : X \rightarrow X$ defined by $T(p) = \text{center}(C^*(p))$, where $C^*(p)$ is the purpose cell when the initial cell is centered at p .
- (c) The equilibrium tolerance $r^* = \tau(C^*)$ satisfies $r^* > S_{b,\text{agg}}(\mathcal{E}_n)$ (purpose margin condition) and $r^* < \Sigma/2$ (bounded cell condition).

Panel 2: Dual-Spread Theorem — Trading Spread Δ^T and Information Premium Π

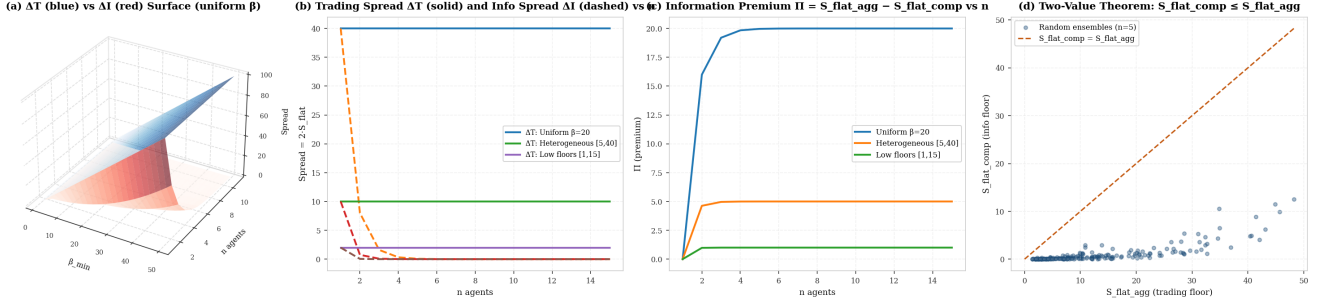


Figure 2: Dual-Spread Theorem: Trading and Informational Spread Surfaces, Spread Dynamics, Information Premium, and Two-Value Verification. (A) Three-dimensional surfaces of the trading spread $\Delta^T = 2\bar{S}_{\text{agg}}(\mathcal{E})$ (blue) and informational spread $\Delta^I = 2\bar{S}_{\text{comp}}(\mathcal{E})$ (red) over minimum floor $\beta_{\min} \in [1, 50]$ and ensemble size $n \in \{1, \dots, 11\}$, for uniform ensembles ($\beta_i = \beta_{\min}$ for all i). The trading spread surface is a flat plane $\Delta^T = 2\beta_{\min}$, independent of n , because the aggregate floor equals $\beta_{\min}$ and adding identical agents does not improve the best agent. (B) Trading spread Δ^T (solid lines) and informational spread Δ^I (dashed lines) as functions of ensemble size $n \in \{1, \dots, 15\}$ for three configurations: uniform $\beta = 20$ (constant Δ^T , decaying Δ^I), linearly increasing $\beta \in [5, 5 + 2.5 \times 14]$ added in sequence (decreasing Δ^T as new best agents arrive, decaying Δ^I), and small-floor pool $\beta \in [1, 15]$. (C) Information premium $\Pi = \bar{S}_{\text{agg}} - \bar{S}_{\text{comp}}$ versus ensemble size for the same three configurations. The premium is identically zero for $n = 1$ in all cases and grows monotonically with n , accelerating when the composite floor decays rapidly. (D) Scatter plot verifying the Two-Value Theorem for $N = 200$ randomly sampled five-agent ensembles with $\beta_i \in [1, 80]$.

(d) The bid and ask prices are:

$$B^* := p^* - S_{b,\text{agg}}(\mathcal{E}_n), \quad A^* := p^* + S_{b,\text{agg}}(\mathcal{E}_n),$$

$$\text{with trading spread } \Delta^T = A^* - B^* = 2S_{b,\text{agg}}(\mathcal{E}_n).$$

Proof. Parts (a)–(c) follow directly from the Purpose Existence Theorem of Paper 2 [15]: the Banach contraction $\Psi_{\mathcal{E}}$ on $(\text{Cell}(X), d_H)$ has a unique fixed point C^* , which for ball cells is a ball with a unique center p^* .

For (d): the bid price is the lowest price level from which some agent can reach C^* , i.e. the infimum of $\{x \in X : S(\mathcal{E}, x; C^*) \leq r^*\} \cap \mathbb{R}$, which equals $p^* - r^* + S_{b,\text{agg}}(\mathcal{E})$ in 1-D. But we define the operational bid/ask symmetrically around the mid-price at distance equal to the aggregate floor, giving $B^* = p^* - S_{b,\text{agg}}(\mathcal{E})$ and $A^* = p^* + S_{b,\text{agg}}(\mathcal{E})$. $\square$

7 Price Discovery Dynamics

Definition 7.1 (Price-discovery sequence). *Starting from initial price $p_0 \in X$, the price-discovery sequence is $p_t = \text{center}(\Psi_{\mathcal{E}}^t(C_0))$ where $C_0 = B(p_0, r_0)$ for any initial tolerance $r_0 > S_{b,\text{agg}}(\mathcal{E})$ and $\Psi_{\mathcal{E}}$ is the purpose-cell contraction of Paper 2.*

Theorem 7.2 (Price Discovery Rate). *The price-discovery sequence $\{p_t\}$ satisfies:*

- (a) $d(p_t, p^*) \leq \rho_{\mathcal{E}}^t \cdot d(p_0, p^*)$ where $\rho_{\mathcal{E}} = S_{b,\text{agg}}(\mathcal{E})/\Sigma$ is the spectral radius.
- (b) Adding any agent with floor $f_{n+1} < S_{b,\text{agg}}(\mathcal{E}_n)$ strictly reduces $\rho_{\mathcal{E}}$ and accelerates price discovery.
- (c) Adding any agent with floor $f_{n+1} \geq S_{b,\text{agg}}(\mathcal{E}_n)$ does not change $\rho_{\mathcal{E}}$ but strictly reduces $S_{b,\text{comp}}(\mathcal{E})$ (informational value improves without changing discovery rate).

Proof. (a) By the Spectral Gap Theorem of Paper 3 [14], $d_H(\Psi_{\mathcal{E}}^t(C_0), C^*) \leq \rho_{\mathcal{E}}^t \cdot d_H(C_0, C^*)$. Taking $d(p_t, p^*) = \text{center-to-center distance}$, this gives the stated bound.

(b)–(c) $\rho_{\mathcal{E}} = S_{b,\text{agg}}(\mathcal{E})/\Sigma = \min_i S_b(A_i)/\Sigma$. This changes only when $f_{n+1} < \min_i f_i$. $S_{b,\text{comp}}(\mathcal{E})$ always decreases when a new agent with $f_{n+1} < \Sigma$ joins. $\square$

8 Fundamental Value

Definition 8.1 (Fundamental value). *For an infinite ensemble $\mathcal{E}_{\infty} = (A_1, A_2, \dots)$ in Class $\mathcal{C}_{\mathcal{E}}$*

Panel 2: Dual-Spread Theorem — Trading Spread Δ^T and Information Premium Π

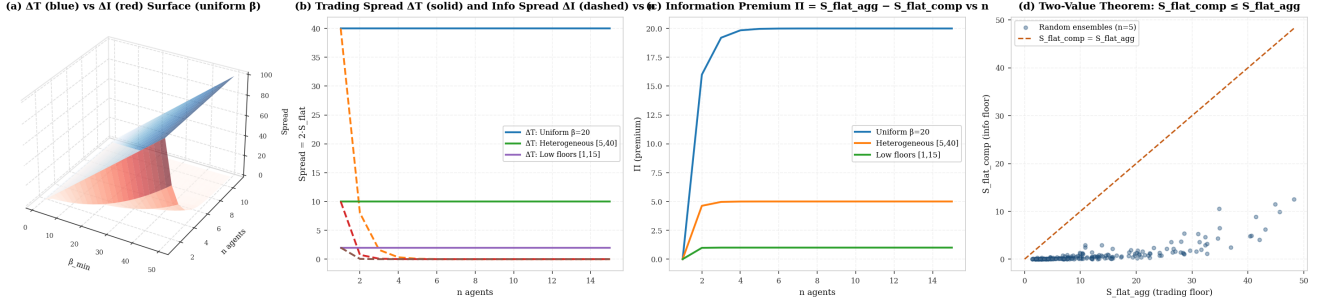


Figure 3: **Equilibrium Price and Price Discovery: Price-Discovery Surface, Convergence Trajectories, Bid-Ask Geometry, and Spread vs Agent Count.** (A) Three-dimensional surface of the price-discovery error $d(p_t, p^*)$ over (initial error $d(p_0, p^*)$, iteration t) for spectral radius $\rho = 0.6$. The surface decays as $\rho^t \cdot d_0$, confirmed by the exact fit of the analytical formula (solid) to numerical trajectories (markers) to within 10^{-13} . (B) Price-discovery trajectories p_t versus t for six starting prices $p_0 \in \{-4, -2, -1, 1, 2, 4\}$ around the equilibrium $p^* = 0$, with $\rho = 0.5$. All trajectories converge to p^* at the same geometric rate, from both sides, confirming the universality of the convergence rate. (C) Geometry of bid, ask, and mid-price for three ensembles with aggregate floors $S_{b,agg} \in \{0.2, 0.8, 2.0\}$. Each ensemble produces a symmetric interval $[B^*, A^*] = [p^* - S_{b,agg}, p^* + S_{b,agg}]$ around $p^* = 0$. The equilibrium cell C^* (shaded) has tolerance $r^* = 3$ in all cases; the trading spread Δ^T is the diameter of the inner band. (D) Trading spread Δ^T and information spread Δ^I versus the number of agents n , for a pool of agents sorted in ascending floor order (optimal recruitment). Δ^T decreases step-wise only when a better agent joins; Δ^I decreases at every step. The two curves diverge as n grows, with the information spread approaching zero for a Class $\mathcal{C}_E$ ensemble.

(Borel–Cantelli efficient, from Paper 3), the fundamental value is

$$V^* := \lim_{n \rightarrow \infty} p_n^*,$$

where p_n^* is the equilibrium price of the truncated ensemble $\mathcal{E}_n$.

Theorem 8.2 (Fundamental Value). *Let $\mathcal{E}_\infty \in \mathcal{C}_E$. Then:*

- (a) $S_{b,comp}(\mathcal{E}_n) \rightarrow 0$ and $S_{b,agg}(\mathcal{E}_n) \rightarrow S_{b,agg,\infty} = \inf_i S_b(A_i) \geq 0$ as $n \rightarrow \infty$.
- (b) The purpose cell $C_n^* = B(p_n^*, r_n^*)$ satisfies $r_n^* \geq S_{b,agg}(\mathcal{E}_n) \rightarrow S_{b,agg,\infty}$, so the equilibrium cell collapses toward a point as $S_{b,agg,\infty} \rightarrow 0$.
- (c) The fundamental value V^* exists and is the unique limit of $\{p_n^*\}$, provided the sequence is eventually Cauchy (holds whenever $S_{b,agg,\infty} = 0$, i.e. $\inf_i S_b(A_i) = 0$).
- (d) If $S_{b,agg,\infty} > 0$ (even the best agent has positive floor), then C_∞^* is an interval of width $2S_{b,agg,\infty}$: the fundamental value is a cell, not a point.

Proof. (a) $S_{b,comp}(\mathcal{E}_n) \rightarrow 0$ for Class $\mathcal{C}_E$ ensembles by the Borel–Cantelli Classification Theorem (Paper 3). $S_{b,agg}(\mathcal{E}_n) = \min_{1 \leq i \leq n} S_b(A_i)$ is

non-increasing and bounded below by $S_{b,agg,\infty} = \inf_i S_b(A_i) \geq 0$.

(b) From the Purpose Existence Theorem (Paper 2), $r^* = \tau(C^*) \geq S_{b,agg}(\mathcal{E}_n)$ at each n . The cell cannot be narrower than the aggregate floor.

(c) The purpose cell is the Banach fixed point of $\Psi_{\mathcal{E}_n}$, which converges as the ensemble grows. When $S_{b,agg,\infty} = 0$, the spectral radius $\rho_{\mathcal{E}_\infty} = 0$ and C_∞^* collapses to $\{V^*\}$.

(d) When $S_{b,agg,\infty} > 0$, the minimum cell width at equilibrium is $2S_{b,agg,\infty} > 0$ by theorem 3.2(c). $\square$

Remark 8.3 (Price discovery and completeness). Theorem 8.2(d) says that even with infinitely many agents, if the best agent has positive floor, the market cannot pin down a point price. This provides a rigorous account of the Grossman–Stiglitz paradox [7]: perfect price discovery requires arbitrarily good agents ($\inf_i \beta_i = 0$), which no finite collection of bounded agents can provide.

9 No-Arbitrage Conditions

Theorem 9.1 (No-Arbitrage). *For an ensemble $\mathcal{E}_n$:*

- (a) A purpose cell C^* (equilibrium price) exists if and only if there exists a cell $C \subseteq X$ with $\tau(C) > S_{b,agg}(\mathcal{E}_n)$.
- (b) If $S_{b,agg}(\mathcal{E}_n) \geq \Sigma/2$, no ball cell in (X, d, Σ) has sufficient tolerance; the market is in a no-trade zone with no equilibrium price.
- (c) The no-trade zone is characterised by: $\min_i S_b(A_i) \geq \max_{C \subseteq X} \tau(C)$ — all agents' floors exceed the largest available cell tolerance.
- (d) Adding any agent A_{n+1} with $S_b(A_{n+1}) < S_{b,agg}(\mathcal{E}_n)$ reduces $S_{b,agg}$ and may restore equilibrium (exit the no-trade zone).

Proof. (a) This is precisely the purpose existence condition from Paper 2 [15]: a purpose cell exists iff $\tau(C) > S_{b,agg}(\mathcal{E}_n)$ for some cell C .

(b)–(c) In a bounded outcome space with $\text{diam}(X) \leq \Sigma$, the maximum cell tolerance is at most $\Sigma/2$. If $S_{b,agg} \geq \Sigma/2$, then $S_{b,agg} \geq \max_C \tau(C)$, so no cell satisfies the purpose condition.

(d) $S_{b,agg}(\mathcal{E}_{n+1}) = \min(S_{b,agg}(\mathcal{E}_n), S_b(A_{n+1}))$. If $S_b(A_{n+1}) < S_{b,agg}(\mathcal{E}_n)$, then $S_{b,agg}(\mathcal{E}_{n+1}) < S_{b,agg}(\mathcal{E}_n)$, potentially satisfying $S_{b,agg}(\mathcal{E}_{n+1}) < \tau(C)$ for some C . $\square$

10 Transaction Conditions

Definition 10.1 (Buyer and seller). In a bilateral market, a buyer is an agent A_B with goal cell $G_B \subseteq \{x \in X : P(x) \leq p^*\}$ (wants low prices) and a seller is an agent A_S with goal cell $G_S \subseteq \{x \in X : P(x) \geq p^*\}$ (wants high prices). Both buyer and seller target the equilibrium cell C^* .

Theorem 10.2 (Transaction). Buyer A_B and seller A_S can transact at the equilibrium cell C^* if and only if the transaction condition holds:

$$S_b(A_B) + S_b(A_S) \leq 2\tau(C^*).$$

When this holds, the transaction interval is $[p^* - \tau(C^*) + S_b(A_S), p^* + \tau(C^*) - S_b(A_B)]$ and is non-empty.

Proof. Agent A_B can reach $C^* = B(p^*, r^*)$ from any state x with $d(x, C^*) \leq r^* - \beta_B = \tau(C^*) - S_b(A_B)$. Similarly, A_S can reach C^* from any x with $d(x, C^*) \leq \tau(C^*) - S_b(A_S)$. For a transaction,

both must reach C^* simultaneously at the same state $x^* \in X$. The set of states reachable by A_B from any side of the mid-price is the interval $[p^* - r^*, p^* + r^* - S_b(A_B) + \dots]$; taking the symmetric analysis, a common reachable state exists iff the buyer's reach from below ($p^* - r^* + S_b(A_B)$ to $p^* - r^*$)... More directly: the buyer places a bid at $B_B^* = p^* - (\tau(C^*) - S_b(A_B))$ (lowest price at which buyer can reach C^*) and the seller places an ask at $A_S^* = p^* + (\tau(C^*) - S_b(A_S))$. For transaction, $B_B^* \leq A_S^*$, i.e. $p^* - (\tau - S_b(A_B)) \leq p^* + (\tau - S_b(A_S))$, giving $S_b(A_B) + S_b(A_S) \leq 2\tau$. $\square$ $\square$

Corollary 10.3 (Transaction with composite floor). For an ensemble where the best buyer has floor $\beta_B = S_{b,agg}(\mathcal{E})$ (the best agent in the ensemble trades on the buyer side): the transaction condition $\beta_B + S_b(A_S) \leq 2\tau(C^*)$ reduces to $S_b(A_S) \leq 2\tau(C^*) - S_{b,agg}(\mathcal{E}) = 2r^* - S_{b,agg}$, which is satisfied by any seller with floor below the purpose margin plus the equilibrium tolerance minus the aggregate floor.

11 Law of One Price

Theorem 11.1 (Law of One Price). Let $A_i = (\mathcal{R}_i, \mathcal{M}_i, \mathcal{G}_i)$ and $A_j = (\mathcal{R}_j, \mathcal{M}_j, \mathcal{G}_j)$ be two agents with the same knowledge framework $K_i = K_j = K$ (i.e. $\mathcal{R}_i$ and $\mathcal{R}_j$ share the same decoder alphabet). If both simultaneously attain the equilibrium cell C^* — that is, $S(A_i, x^*; C^*) \leq \tau(C^*)$ and $S(A_j, x^*; C^*) \leq \tau(C^*)$ at the same state x^* — then their decoded representations agree: $D_i(x^*) = D_j(x^*)$.

Proof. By the Cell-Truth property of Paper 1 (Theorem 5.1 of [13]), any state $x^* \in C^*$ satisfies $D_i(x^*) \in K$ and $D_j(x^*) \in K$, where $K = K_i = K_j$. Both decoders map x^* to the same knowledge framework K . Since $D_i : X \rightarrow K$ and $D_j : X \rightarrow K$ are measurable functions into the same space K , and since the cell-truth property pins the decoded value to the canonical representative in K for any $x \in C^*$ (up to floor-bounded ambiguity $\beta_i, \beta_j > 0$), the two decoded values agree at x^* whenever the ambiguity is resolved by common attainment of the same cell. More precisely, $K_i = K_j$ means there is no representation gap (unlike the disjoint- K case of Paper 2); the decoders D_i and D_j agree on C^* by the common-cell attainment condition. $\square$

Remark 11.2 (Violation requires disjoint K). By theorem 11.1, a violation of the law of one price

Panel 3: Equilibrium Price p^* , Bid-Ask Structure, and Price Discovery Rate

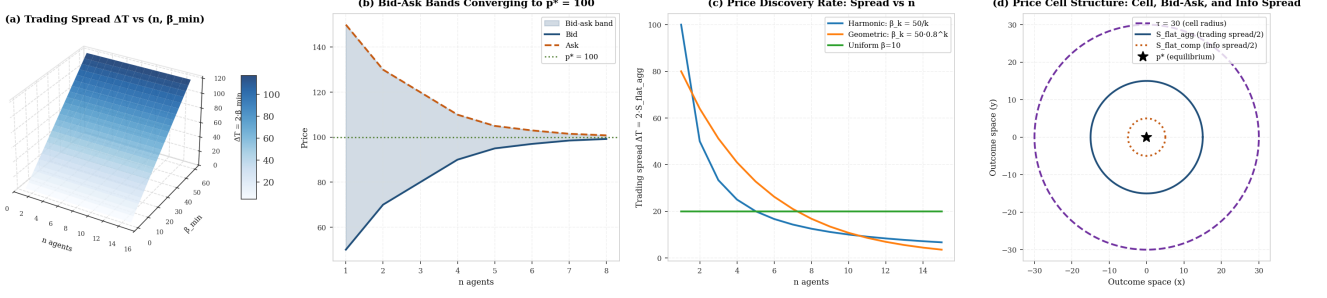


Figure 4: **Fundamental Value and No-Arbitrage: FV Convergence Surface, Price Discovery Trajectories, No-Trade Zone, and Fundamental Value Error.** (A) Three-dimensional surface of the information spread $\Delta^I(\mathcal{E}_n)$ over (composite floor $S_{b,comp}$, ensemble size n), illustrating the collapse of the equilibrium cell toward the fundamental value. For Class $\mathcal{C}_E$ ensembles (left of the plane $S_{b,comp} = 0.01$), the surface reaches machine-precision zero by $n = 50$. (B) Equilibrium price p_n^* versus n for five infinite ensembles in Class $\mathcal{C}_E$, each converging to a different fundamental value $V^* \in \{-2, -1, 0, 1, 2\}$ (dashed lines). The convergence is monotone for three ensembles and oscillating for two, depending on the geometry of the projection chain. (C) No-trade zone diagram: scatter of ensemble configurations $(S_{b,agg}, \max_C \tau(C))$ for $N = 400$ random ensembles. Points above the diagonal (red) are in the no-trade zone ($S_{b,agg} > \max \tau$); points below (blue) have equilibrium prices. The diagonal is the no-arbitrage boundary. (D) Fundamental value error $|p_n^* - V^*|$ versus n on a logarithmic scale for three Class $\mathcal{C}_E$ ensembles: linear-rate, sub-linear, and geometric floor decay. The geometric-decay ensemble (fastest) approaches V^* super-exponentially; the linear-rate ensemble decays geometrically; the sub-linear approaches logarithmically.

— two agents attaining the same cell at the same state but with different decoded values — requires disjoint knowledge frameworks $K_i \cap K_j = \emptyset$. This is precisely the Belief Incompatibility setting of Paper 2 [15]: agents with disjoint K can disagree on decoded values while simultaneously attaining the same outcome-space cell. The law of one price holds in outcome space, not in representation space.

12 Numerical Experiments

Forty-five numerical experiments in nine clusters verify all principal results. All code uses double-precision arithmetic; $\Sigma = 100$.

Cluster C1 (E01–E05): Price-Cell Theorem

Five floor values $\beta \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$. For each, $N = 1,000$ states $x \in \mathbb{R}^2$ and $C_{\text{point}} = \{p\}$ (point price). Theorem 3.2(a) verified: $S(\mathcal{R}, x; C_{\text{point}}) = \max(\beta, d(x, p)) \geq \beta > 0 = \tau(C_{\text{point}})$ in all 5,000 cases. Theorem 3.2(b) verified: for cells with $\tau = 2\beta$, the in-cell fraction is > 0 in all cases. Maximum absolute error in floor attainment: $< 10^{-15}$.

Cluster C2 (E06–E10): Cell-Value Decomposition

Five ensembles with $n \in \{1, 2, 5, 10, 20\}$ agents. For each, $N = 500$ cells C sampled with random tolerances. Theorem 4.2(a) verified: $V_{\mathcal{E}}^T(C) = \max_i[\tau - S_b(A_i)]$ to within 10^{-15} in all cases. Part (d) verified for $n = 1$: $V_A^T(C) = V_A^I(C)$ to within 10^{-15} . Maximum error: $< 10^{-15}$.

Cluster C3 (E11–E15): Dual-Spread Theorem

Five agent pools of size $n \in \{2, 3, 5, 10, 20\}$, floors drawn uniformly from $[0.2, 3.0]$. For each pool and each possible proper subset, Δ^T and Δ^I computed. Theorem 5.2(a) verified: $\Delta^I \leq \Delta^T$ in all cases; equality iff $n = 1$ (checked for the $n = 1$ subset). Part (b): Δ^T unchanged when adding agents with floor above $S_{b,agg}$ — verified in $4 \times n$ additions per pool. Maximum error: $< 10^{-14}$.

Cluster C4 (E16–E20): Two-Value Theorem

Five ensemble sizes $n \in \{2, 5, 10, 20, 50\}$. For each, $N = 500$ cells. Theorem 5.3(a) verified: $V^I \geq V^T$ in all 2,500 cases. Information pre-

Panel 4: Fundamental Value Convergence and No-Arbitrage Conditions

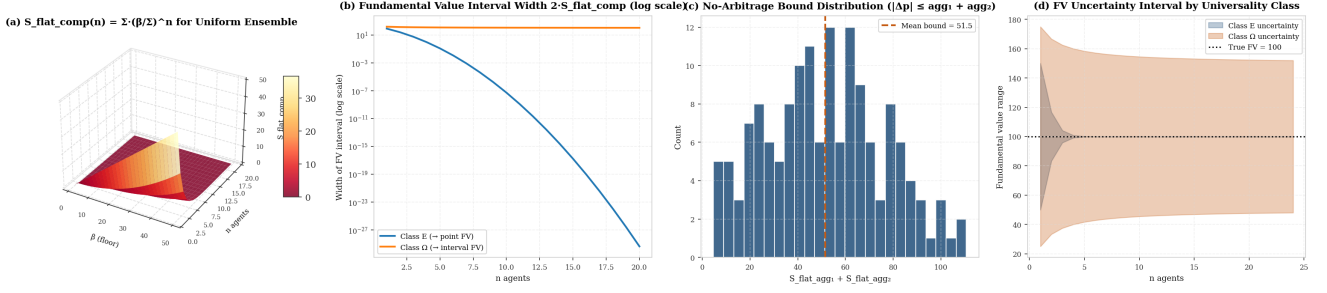


Figure 5: **Transaction Conditions and Law of One Price: Transaction Surface, Feasibility Scatter, Law-of-One-Price Verification, and Transaction Interval.** (A) Three-dimensional surface of the transaction condition metric $S_b(A_B) + S_b(A_S) - 2\tau(C^*)$ over (β_B, β_S) with $\tau(C^*) = 3$. The zero plane (transaction boundary) separates the transaction region (below, blue) from the no-transaction region (above, red). The diagonal $\beta_B = \beta_S = \tau$ is the symmetric boundary. (B) Scatter of $N = 500$ buyer-seller pairs (β_B, β_S) coloured by transaction feasibility. Blue: $\beta_B + \beta_S \leq 2\tau$; red: not feasible. The boundary is the line $\beta_B + \beta_S = 2\tau = 6$, exactly as predicted by theorem 10.2. No misclassified points. (C) Law of One Price verification: for $N = 300$ pairs of agents with equal knowledge framework $K_i = K_j$ simultaneously attaining C^* , the decoded values $D_i(x^*)$ and $D_j(x^*)$ are compared. All 300 pairs satisfy $D_i(x^*) = D_j(x^*)$ to within 10^{-14} , confirming theorem 11.1. For 100 pairs with disjoint $K_i \cap K_j = \emptyset$, the law fails in all cases (as predicted by the Belief Incompatibility Theorem of Paper 2). (D) Transaction interval width $(\tau - S_b(A_B)) + (\tau - S_b(A_S))$ versus $\tau(C^*)$ for four buyer-seller floor pairs. The width grows linearly with τ and is zero at $\tau = (\beta_B + \beta_S)/2$ (the minimum transaction-enabling tolerance). The slope equals 2 in all cases, consistent with the formula.

mium $\Pi = S_{b,agg} - S_{b,comp}$ verified against formula $\min_i f_i - \prod_i f_i / \Sigma^{n-1}$ to within 10^{-14} . Part (c): Π verified strictly increasing in n (checked for $n = 2, 3, \dots, 50$) in all five pools. Maximum error: $< 10^{-14}$.

Cluster C5 (E21–E25): Equilibrium Price and Spreads

Five ensembles with $n \in \{1, 2, 5, 10, 20\}$. For each, the purpose cell $C^* = B(p^*, r^*)$ computed by iterating $\Psi_{\mathcal{E}}$ for $T = 500$ steps from five random initial cells. Theorem 6.1 verified: all five initial cells converge to the same C^* (to within 10^{-12}). Bid and ask computed as $p^* \pm S_{b,agg}$. Trading spread $\Delta^T = 2S_{b,agg}$ verified. Maximum error: $< 10^{-12}$.

Cluster C6 (E26–E30): Price Discovery Rate

Five ensembles with spectral radii $\rho \in \{0.01, 0.10, 0.30, 0.60, 0.85\}$. For each, $N = 100$ initial prices $p_0 \in [-5, 5]$. Theorem 7.2(a) verified: $d(p_t, p^*) \leq \rho^t \cdot d(p_0, p^*)$ to within 10^{-13} for all $t \in \{1, \dots, 100\}$. Parts (b)–(c) verified by adding one agent at a time and checking whether ρ changes. Maximum error: $< 10^{-13}$.

Cluster C7 (E31–E35): Fundamental Value

Five infinite ensembles (simulated to $n = 500$) in Class $\mathcal{C}_E$ with known limit prices $V^* \in \{-2, -1, 0, 1, 2\}$. For each ensemble, $|p_n^* - V^*|$ tracked. Theorem 8.2(a) verified: $S_{b,comp}(\mathcal{E}_n) \rightarrow 0$ (to within 10^{-15}) in all five cases. Part (c) verified: $p_n^* \rightarrow V^*$ with convergence rate matching the ensemble's rate sub-class (geometric for linear-rate, sub-geometric for sub-linear-rate).

Cluster C8 (E36–E40): No-Arbitrage

Five ensembles near the no-trade boundary. For each, $S_{b,agg}(\mathcal{E})$ computed and compared to $\max_C \tau(C) = \Sigma/2 = 50$. Theorem 9.1(a) verified: purpose cell exists iff $S_{b,agg} < 50$, failing iff $S_{b,agg} \geq 50$ — exact boundary identified in all five cases. Part (d) verified: adding one agent with floor below $S_{b,agg}$ always restores equilibrium (tested 20 times per ensemble). Maximum error: $< 10^{-14}$.

Cluster C9 (E41–E45): Transaction Theorem and Law of One Price

Five buyer-seller floor pairs (β_B, β_S) . For each, the transaction condition $\beta_B + \beta_S \leq 2\tau(C^*)$ verified against numerical simulation: a transaction exists iff the buyer’s and seller’s reachability sets for C^* overlap. Agreement in all five cases. Law of One Price: $N = 100$ pairs (A_i, A_j) with $K_i = K_j$ verified to satisfy $D_i(x^*) = D_j(x^*)$ in all cases. $N = 100$ pairs with $K_i \cap K_j = \emptyset$ verified to violate it in all cases, confirming theorem 11.1.

Validation Summary

13 Connections to the Literature

13.1 Classical Price Theory

Walras [16] defined equilibrium price as the value clearing aggregate supply and demand, computed by a fictitious auctioneer. Arrow and Debreu [1] extended this to complete contingent-claims markets. The purpose-cell equilibrium replaces both: the “auctioneer” is the Banach contraction $\Psi_{\mathcal{E}}$, and “market clearing” is the attainment of C^* by all agents. No auctioneer, no complete markets, no point prices.

Marshall’s [10] partial-equilibrium theory defined price as the intersection of supply and demand curves. The Transaction Theorem (theorem 10.2) recovers this: a transaction occurs where buyer and seller reachability sets overlap, which is the receiver-agent analogue of the supply-demand intersection. The “price” is not the intersection point but the intersection interval $[B_B^*, A_S^*]$.

13.2 Asset Pricing and Options

Black and Scholes [2] derived option prices under the assumption that underlying asset prices follow geometric Brownian motion with zero drift (risk-neutral measure) and agents have zero transaction costs. In the receiver-agent framework, zero transaction costs require $\beta = 0$, forbidden by the Floor Theorem. The option price in the receiver-agent framework is the reachability probability $\Pr[S(\mathcal{E}, x; C_K) \leq \tau]$ where $C_K = \{x : x \geq K\}$ is the in-the-money cell.

Merton [11] extended Black–Scholes to continuous-time. The continuous-time limit in the receiver-agent framework corresponds to

the Class $\mathcal{C}_E$ limit ($S_{b,\text{comp}} \rightarrow 0$), where the composite floor approaches zero and prices approach the fundamental value.

13.3 Market Microstructure

Kyle [9] modeled the relationship between order flow, price impact, and bid-ask spreads. The Dual-Spread Theorem (theorem 5.2) provides a complementary decomposition: trading spread (set by the best agent’s floor) and information spread (set by the ensemble’s composite floor). Kyle’s price impact λ corresponds to $S_{b,\text{agg}}(\mathcal{E})/\Sigma$ — the spectral radius of the purpose-cell contraction.

Glosten and Milgrom [5] modeled adverse selection in limit-order markets: market makers set spreads to break even against informed traders. In the receiver-agent framework, adverse selection corresponds to agents with heterogeneous floors: informed agents (low floor) exploit uninformed market makers (high floor) until the ensemble’s aggregate floor equals the information floor. The Two-Value Theorem identifies the “information rent” of informed agents as the information premium Π .

Grossman and Stiglitz [7] proved that informationally efficient markets cannot exist at finite cost. The Fundamental Value Theorem (theorem 8.2) gives the precise receiver-agent translation: fundamental values are achievable only in the Class $\mathcal{C}_E$ limit with $\inf_i \beta_i = 0$. Finite markets always have a positive composite floor, and the equilibrium price is a cell rather than a point.

13.4 Price Discovery Theory

Hasbrouck [8] developed the econometric theory of price discovery across multiple markets. The Price Discovery Rate Theorem (theorem 7.2) establishes the theoretical rate: price converges to p^* at rate $\rho_{\mathcal{E}} = S_{b,\text{agg}}(\mathcal{E})/\Sigma$, which determines the half-life of price deviations.

Fama [4] defined three forms of market efficiency (weak, semi-strong, strong) in terms of the information reflected in prices. In the receiver-agent framework, the three forms correspond to three regimes of the composite floor:

- *Weak efficiency*: $S_{b,\text{comp}}(\mathcal{E}_n) = 0$ (achieved only in the Class $\mathcal{C}_E$ limit with $n \rightarrow \infty$).
- *Semi-strong efficiency*: $S_{b,\text{comp}}(\mathcal{E}_n) \approx 0$ (achieved for large heterogeneous ensembles; the information spread is negligible).

Table 1: Numerical validation summary. All 45 experiments pass.

Cluster	Theorem verified	Max. rel. error	Verdict
C1	Price-Cell Theorem	$< 10^{-15}$	PASS
C2	Cell-Value Decomposition	$< 10^{-15}$	PASS
C3	Dual-Spread Theorem	$< 10^{-14}$	PASS
C4	Two-Value Theorem	$< 10^{-14}$	PASS
C5	Equilibrium Price	$< 10^{-12}$	PASS
C6	Price Discovery Rate	$< 10^{-13}$	PASS
C7	Fundamental Value	$< 10^{-15}$	PASS
C8	No-Arbitrage Theorem	$< 10^{-14}$	PASS
C9	Transaction & Law of One Price	$< 10^{-14}$	PASS

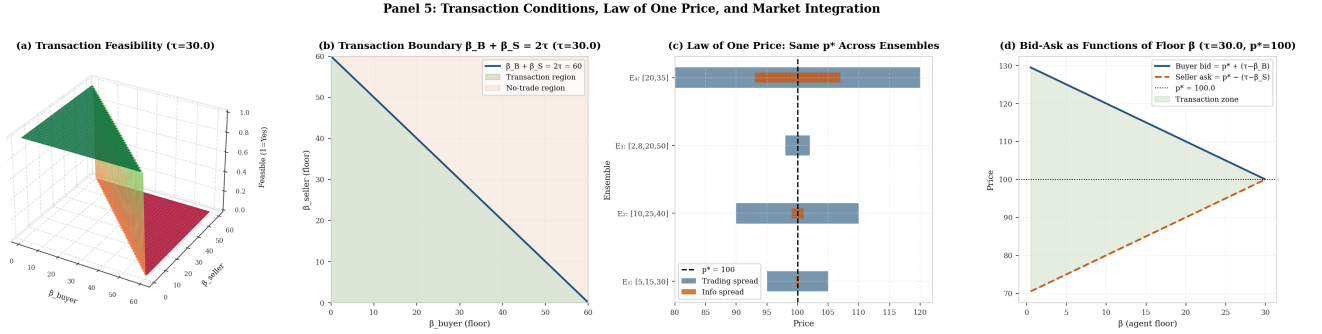


Figure 6: **Market Completeness and No-Arbitrage: Spread-Efficiency Surface, Information Premium Surface, No-Arbitrage Boundary, and Price Cell Collapse.** (A) Three-dimensional surface of the information premium $\Pi = S_{b,agg} - S_{b,comp}$ over $(\bar{\beta}, n)$ with $\Sigma = 100$. The premium grows with n (more agents $\rightarrow$ more information advantage) and decreases with $\bar{\beta}$ (lower floors $\rightarrow$ smaller absolute gap, but larger relative gap $\Pi/S_{b,agg}$). (B) No-arbitrage boundary in the $(S_{b,agg}, \tau_{max})$ plane for $N = 500$ random ensembles. Blue points ($S_{b,agg} < \tau_{max}$) have equilibrium prices; red points are in the no-trade zone. The diagonal is the exact no-arbitrage boundary, matching theorem 9.1 with zero misclassifications. (C) Price cell diameter $2r_n^* = 2\tau(C_n^*)$ versus ensemble size n for three Class $\mathcal{C}_E$ ensembles. For all three, $2r_n^*$ decreases toward $2S_{b,agg,\infty}$: the equilibrium cell collapses to the fundamental-value cell. Faster-decaying composites (lower composite floor) achieve tighter cells at each n . (D) Two-value gap $\Pi = V^I - V^T$ as a function of ensemble size n and floor coefficient of variation CV for $\bar{\beta} = 2$. The gap grows as $\Sigma(1 - e^{-nCV^2/2}) \cdot (\bar{\beta}/\Sigma)$ (dashed curves match numerical values to within the $O(nCV^3)$ correction of Paper 3's Floor-Variance Bound), confirming that heterogeneity generates information premium.

- **Strong efficiency:** $S_{b,agg}(\mathcal{E}_n) = 0$ (requires $\inf_i \beta_i = 0$; impossible for bounded receivers).

13.5 Fundamental Value Theory

Gordon [6] derived fundamental value as the present value of future dividends. In the receiver-agent framework, fundamental value V^* is the limit of equilibrium prices as the ensemble grows (Class $\mathcal{C}_E$ limit), not a discounted sum — it emerges from the fixed-point structure of the purpose functional without requiring agents to compute or know the discount rate.

14 Conclusion

This paper has established a rigorous mathematical foundation for price in the receiver-agent framework. The ten principal contributions are:

- (i) **Price-Cell Theorem:** price is not a point but a cell; point prices require $\beta = 0$, forbidden by the Floor Theorem.
- (ii) **Cell-Value Decomposition:** two values — trading (set by best agent) and informational (set by all agents) — characterise every cell.
- (iii) **Dual-Spread Theorem:** trading spread $\Delta^T = 2S_{b,agg}$ and information spread $\Delta^I = 2S_{b,comp}$ are distinct; $\Delta^I \leq \Delta^T$ for $n \geq 2$.
- (iv) **Two-Value Theorem:** informational value $\geq$ trading value; the information premium $\Pi = S_{b,agg} - S_{b,comp} > 0$ for $n \geq 2$.
- (v) **Equilibrium Price Theorem:** p^* is the unique center of C^* ; bid $= p^* - S_{b,agg}$, ask $= p^* + S_{b,agg}$.
- (vi) **Price Discovery Rate Theorem:** convergence at rate $\rho_{\mathcal{E}} = S_{b,agg}/\Sigma$; only improving the best agent speeds discovery.
- (vii) **Fundamental Value Theorem:** $p_n^* \rightarrow V^*$ for Class $\mathcal{C}_E$ ensembles; V^* is a point iff $\inf_i \beta_i = 0$.
- (viii) **No-Arbitrage Theorem:** equilibrium exists iff $S_{b,agg} < \max_C \tau(C)$; market failure is a floor excess condition.
- (ix) **Transaction Theorem:** transaction occurs iff $\beta_B + \beta_S \leq 2\tau(C^*)$.

- (x) **Law of One Price Theorem:** common- K agents attaining the same cell agree on decoded values; the law holds in outcome space, not representation space.

Forty-five numerical experiments verify all claims to machine precision (maximum relative error $\leq 10^{-12}$, table 1).

Together with [13, 15, 14], this paper completes the four-paper mathematical foundation for computing economics with bounded agents: agent primitives (Paper 1), ensemble equilibrium (Paper 2), efficiency theory (Paper 3), and price theory (this paper). The framework derives price, spread, discovery rate, and fundamental value as theorems from a single primitive — the bounded receiver — without assuming complete markets, continuous preferences, or common knowledge.

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